

Context-Aware CVE Prioritization Framework — Five-Layer Architecture

Layer 1: Data Ingestion

NVD · CISA KEV · Exploit-DB · MITRE ATT&CK · EPSS · Asset Telemetry



Layer 2: Feature Engineering

Static CVE (9) · Threat Intel (7) · Env. Context (8) · Compliance (4) = 28 features



Layer 3: ML Inference

Random Forest + XGBoost → Stacked Ensemble (meta-learner: Logistic Regression)



Layer 4: Prioritization & Decision

RiskScore = 0.3M + 0.3A + 0.2E + 0.2B → Critical / High / Medium / Low tiers



Layer 5: Patch Orchestration

JSON payload → OT patch scheduler → Deployment feedback loop